

Intrinsic-Signal “Do-No-Harm” Gating for a Frozen-Base Soft-Prompt Memory Wrapper

Settings, Metric Definitions, and a Signal–Usefulness Correlation Study (v1.5)

Plan 08 (mem-X line) — internal technical note

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Abstract

v1 of this project showed that a learned K -token soft-prompt memory wrapper around a frozen LLM is a *narrow lossy compressor*: it matches the Gist-Tokens baseline on QuALITY but *negatively transfers* on benchmarks it never trained for (verbatim retrieval, multi-hop QA), sometimes scoring *below* the base model with no wrapper at all. Because the whole point of a frozen base is that adding a module should not *lose* points, v1.5 asks: can we *gate* the wrapper’s intervention — suppress it when it would hurt — using cheap signals computable at inference? This note fixes **all experimental settings** (§1) and **all metric definitions** (§2), both the benchmark-native black-box outcomes and the intrinsic signals (including *per-layer* logit-lens quantities), then reports the first correlation study (§3) identifying *which* intrinsic signals — and at *which layers* — predict the benchmark’s own usefulness measure. The design principle: first learn what correlates with the target, then design the gate around it.

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1 Settings

1.1 Frozen base and the wrapper

The base f_θ is **Qwen3-8B** (hidden size $d = 4096$, 36 decoder layers), *frozen* throughout; only the wrapper trains. The wrapper ϕ_ψ ($\approx 218\text{M}$ trainable params, 2.67% of the base) maintains a memory state $m_t \in \mathbb{R}^{K \times d}$, $K = 32$.

Write side (recurrence). For each context chunk c_t ($t = 1:N$, $N = 6$ chunks of ≤ 256 tokens), a single Perceiver-IO cross-attention+FFN block g_ϕ (heads $h = 8$, FFN mult 2, $L = 1$ layer) produces a candidate, and the wrapper exposes the residual:

$$\Delta m_t = \text{LN}(g_\phi(m_{t-1}, c_t) - m_{t-1}), \quad m_t = m_{t-1} + \alpha_{\text{upd}} \cdot \Delta m_t, \quad (1)$$

with α_{upd} a learned scalar (init 1.0). Δm_t is published on `MemoryState.extra["last_delta"]` for geometry probes.

Read side (combine). At query time the final memory m_T is delivered to the base. The canonical mode is `xattn`: a second cross-attention lets memory re-read the raw chunk pool H_{chunks} , then the refined memory is prepended as a soft prefix:

$$m'_T = m_T + \alpha_{\text{ro}} \cdot \text{LN}(\text{CrossAttn}(m_T, H_{\text{chunks}})), \quad \text{inputs_embeds} = [m'_T; x], \quad (2)$$

where x are the query-token embeddings. Seven combine modes exist (`prefix`, `residual`, `interleave`, `hybrid`, `xattn`, `xattn_residual`, `gated`); only `gated` is unimplemented and is the *target* of v1.5 (§4). No combine mode ever moves a base weight.

Answer head (auxiliary, training only). For categorical training tasks a `Linear(d, n_cls)` head on `pool(m_T)` supplies a direct class-prediction gradient that bypasses the frozen base.

Joint training loss.

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{kl}} \text{KL}(p_{\text{student}} \| p_{\text{teacher}}) + \lambda_{\text{div}} \frac{1}{T-1} \sum_t \cos(\Delta m_t, \Delta m_{t-1}) + \lambda_{\text{ah}} \mathcal{L}_{\text{ah}}, \quad (3)$$

teacher = same base reading the full context (`full_context`).

1.2 Training recipe (locked)

knob	value	knob	value
base	Qwen3-8B (frozen, bf16)	combine	<code>xattn</code>
K (memory tokens)	32	h heads / L layers	8 / 1
train dataset	<code>categorical_niah</code>	N train items	500
N chunks	6	max chunk tokens	256
optimiser	AdamW, lr 3×10^{-5} , const	steps	1800 (SFT)
λ_{kl}	1.0	λ_{div}	0.1
λ_{ah} (answer head)	0.5	seeds	{42,7,11,13,1,5,17,99,23}

1.3 Evaluation protocol

The wrapper is trained *only* on `categorical_niah` and evaluated **zero-shot** on public benchmarks via a `--eval-dataset` flag (wrapper weights never updated on eval data). For every eval item we run, on the same frozen base: (i) the **wrapper** (memory prefix + query); and (ii) **no_context** (query only) — which *is* the base *without* the wrapper, since the wrapper replaces the document with K memory tokens. **no_context** is therefore the do-no-harm reference (§2.2). We also have the standard baselines **full_context** (base reads the full document), **retrieval** (BM25 top-2 chunks), **summary**.

Compute fan-out. The probe (`scripts/probe_v2.py`) trains one wrapper per seed and evaluates all 5 benchmarks with full instrumentation. Seeds are sharded one-per-GPU across 8 GPUs (`sam-dev-test` H200 $\times 1$; `sam-dev-ray` H100 $\times 4$; `sam-dev` H100 $\times 4$) to produce a multi-seed dataset.

1.4 Benchmarks

All are public long-context QA, evaluated zero-shot. “Answer space” drives the native metric (§2.1).

benchmark	source	answer space	n_{eval}	native metric
QuALITY	Pang et al. 2022	4-way MC (A–D)	150	accuracy (likelihood)
MuSR-mm	Sprague et al. 2024	MC (murder myst.)	150	accuracy (likelihood)
RULER-NIAH	Hsieh et al. 2024	6-digit value	150	contains-recall
HotpotQA	Yang et al. 2018	free-form span	150	SQuAD EM/F1
TriviaQA-RC	Joshi et al. 2017	span + aliases	150	EM/F1 (max over aliases)
SQuAD v2	Rajpurkar et al. 2018	span / no-answer	—	EM/F1 (+abstain)
NarrativeQA	Kočiský et al. 2018	open-ended	—	ROUGE-L, BLEU
MS MARCO	Bajaj et al. 2016	free-form	—	ROUGE-L, BLEU-1

2 Metric definitions

We separate three families: (a) the benchmark-*native* black-box *outcome* metrics (§2.1); (b) the *usefulness* labels derived from them, which are what a gate should track (§2.2); and (c) the *intrinsic signals* that are candidate gate *inputs* (§2.3).

2.1 Black-box (benchmark-native) outcome metrics

Principle: score by the benchmark’s own method. Notably, the multiple-choice benchmarks are natively scored by *likelihood*, not free-form generation; our earlier free-generation + letter-parsing under-counted the base (e.g. `full_context` appeared as 0.079 on QuALITY purely as a parsing artifact).

(MC) Length-normalised option log-likelihood. For options $o^{(1)}, \dots, o^{(C)}$ with token sequences $o = (o_1, \dots, o_{|o|})$, score each by the mean log-prob of its continuation given the prompt P :

$$s(o) = \frac{1}{|o|} \sum_{i=1}^{|o|} \log p_{\theta}(o_i \mid P, o_{<i}), \quad \hat{o} = \arg \max_o s(o), \quad \text{acc} = \mathbf{1}[\hat{o} = o^*]. \quad (4)$$

We also record the continuous *margin* `mc_margin` = $s(o^*) - \max_{o \neq o^*} s(o)$ and the gold log-prob $s(o^*)$ — richer (less noisy) targets than binary accuracy at $n=150$.

(span) SQuAD EM / F1. With SQuAD normalisation $\mathcal{N}(\cdot)$ (lowercase; strip punctuation; drop articles {a,an,the}; collapse whitespace):

$$\text{EM} = \mathbf{1}[\mathcal{N}(\hat{y}) = \mathcal{N}(y)], \quad \text{F1} = \frac{2 \text{prec} \cdot \text{rec}}{\text{prec} + \text{rec}}, \quad (5)$$

on the multiset of normalised tokens; over a reference/alias set R the score is $\max_{y \in R}$ (TriviaQA aliases, NarrativeQA refs). HotpotQA sup-fact/joint metrics are *not* applicable (the wrapper predicts no supporting facts). SQuAD v2: a no-answer item scores 1 iff the model abstains; our current harness restricts to answerable items (SQuAD-1.1-style) — an open gap.

(retrieval) RULER contains-recall. $\text{em_value} = \mathbf{1}[y \subseteq \hat{y}]$ (the planted 6-digit value appears in the output; official `acc_match_str_any`).

(generation) ROUGE-L / BLEU. ROUGE-L is the LCS-based F-score $\frac{2P_{\text{lcs}}R_{\text{lcs}}}{P_{\text{lcs}}+R_{\text{lcs}}}$, $P_{\text{lcs}} = \text{LCS}(\hat{y}, y)/|\hat{y}|$, $R_{\text{lcs}} = \text{LCS}(\hat{y}, y)/|y|$; BLEU- n is the brevity-penalised geometric mean of clipped n -gram precisions. Both taken max/over the reference set. ROUGE-L is the reported headline for NarrativeQA and MS MARCO.

2.2 Usefulness labels (the gate’s target) and reference

Let u_w be the wrapper’s native correctness on an item and u_b the reference’s. We define per item:

$$\begin{aligned} \text{correct} &= \mathbf{1}[u_w \geq \tau] \quad (\tau=0.5), \\ \text{noharm} &= \mathbf{1}[u_w \geq u_b], \\ \text{help} &= \mathbf{1}[u_w > \tau \wedge u_b \leq \tau], \quad \text{hurt} = \mathbf{1}[u_w \leq \tau \wedge u_b > \tau], \\ \text{cont} &= \text{continuous target (MC: mc_margin; else native value)}. \end{aligned}$$

Two references. Against `no_context` (base without wrapper) the wrapper is already harmless on 80–100% of items per benchmark (§3); the damaging gap is against `full_context` (compression discards usable context). Hence the gate is best understood as a *compressibility detector*: decide when lossy memory is safe vs. when to fall back to more context. The *ideal* gate maximises `noharm` (never drop below the base) while keeping `help`.

2.3 Intrinsic signals (candidate gate inputs)

All are computed at inference without the gold label. Let the wrapper run produce final-token next-token distribution p_w and per-layer last-token hidden states h_ℓ^w ($\ell = 0:36$); the `no_context` run gives p_0 , h_ℓ^0 .

Final-layer signals. Entropy $H(p) = -\sum_v p_v \log p_v$; confidence $\text{conf} = \max_v p_v$; margin $= p_{(1)} - p_{(2)}$ (top-two gap); divergence $\text{kl_w0} = \text{KL}(p_w \| p_0)$; agreement $\mathbf{1}[\arg \max p_w = \arg \max p_0]$. Reported as `entropy_w`, `conf_w`, `margin_w`, `kl_w0`, `first_tok_agree`.

Per-layer logit-lens signals (the multi-layer probe). Project each layer’s last-token hidden through the frozen final norm and unembedding W_U to get an early “readout” distribution:

$$\pi_\ell = \text{softmax}(W_U \text{LN}_f(h_\ell)). \quad (6)$$

From π_ℓ^w we record `ll_entropy_w_l`, `ll_conf_w_l`, `ll_margin_w_l`; the wrapper-vs-base divergence `ll_kl_wb_l` = $\text{KL}(\pi_\ell^w \parallel \pi_\ell^0)$; and the depth-stabilisation flag `ll_argmax_agree_final_l` = $\mathbf{1}[\arg \max \pi_\ell^w = \arg \max p_w]$. We also record per-layer *hidden drift*

$$\text{drift_l} = \frac{\|h_\ell^w - h_\ell^0\|_2}{\|h_\ell^0\|_2 + \epsilon}, \quad (7)$$

i.e. how much the wrapper perturbs each layer’s residual stream at the answer position. **Rationale (the multi-layer hypothesis):** different depths encode different things — early-layer drift may flag format perturbation (harmful), mid-layer π_ℓ agreement-with-base may flag safe compression, late-layer confidence may flag correctness — so the most predictive signal need not live at the final layer, and a *per-layer* gate can read whichever layer carries the signal.

Memory geometry. `drift_mem` = $\|m_T - m_0\|_F$; `delta_mean` = $\frac{1}{N} \sum_t \|\Delta m_t\|$; `q_chunk_cos` = $\cos(\overline{\text{enc}}(q), \overline{\text{enc}}(c))$ (query vs. chunk mean-pooled encodings; max and mean over chunks).

Ensemble signals (Phase 0 only; not purely intrinsic). `intervene_vs_noctx` = $\mathbf{1}[\hat{y}_w \neq \hat{y}_0]$; `agree_retrieval` = $\mathbf{1}[\hat{y}_w = \hat{y}_{\text{retr}}]$; `n_baselines_agree` = $\#\{\text{baselines whose prediction} = \hat{y}_w\}$; `w_outlen` (generated length). These need running other strategies and so are candidate *ensemble* gate inputs.

2.4 Correlation metrics

For each (signal, layer) we report **AUROC** against a binary label (rank statistic: $\text{AUROC} = \Pr[s_+ > s_-]$; 0.5 = no signal, $|\cdot - 0.5|$ measures discriminative power, < 0.5 means the signal is *inversely* predictive), and **Spearman** ρ against the continuous target. A *per-layer profile* reports, per layer-metric, the layer maximising $|\text{AUROC} - 0.5|$. A planned multivariate logistic probe on the full {metric×layer} vector estimates the achievable gate ceiling.

3 Results and interpretation

Status. The full 9-seed dataset is in flight; tables below are the completed **seed 42** ($n=150/\text{bench}$) plus the Phase-0 free-generation study (5,600 items). Multi-seed bands (pooling $\sim 1,350$ items/bench) will tighten and re-rank these.

3.1 Phase 0 — base rates (free-gen scored, 5,600 items)

bench	u_w	no_context	full_context	help%	hurt%	noharm%
quality	0.193	0.141	0.079	13	8	92
musr_mm	0.492	0.494	0.551	11	12	89
trivia_qa	0.168	0.338	0.565	3	20	80
hotpot_qa	0.086	0.104	0.438	3	5	95
ruler_niah	0.000	0.000	0.994	0	0	100

Read: vs. `no_context` the wrapper is mostly harmless (noharm 80–100%); it *actively* hurts only on TriviaQA (20%). The catastrophe is vs. `full_context` (RULER $0.994 \rightarrow 0$, Hotpot $0.44 \rightarrow 0.09$) — compression discards usable context. So the gate is a compressibility detector, not a corruption filter.

3.2 Seed-42 gate signal sheet (native-MC + per-layer)

bench	native metric	best intrinsic signal	AUROC	predictive layer
hotpot_qa	squad_f1	conf_w	0.753	near-final (34–35)
quality	accuracy (LL)	11_kl_wb@30	0.680	mid-layer 30
trivia_qa	squad_f1	11_conf@34	0.653	layer 34
musr_mm	accuracy (LL)	11_entropy@11	0.582	weak (at chance)
ruler_niah	exact_value	—	—	undefined (0 correct)

Per-layer profile (layer maximising $|\text{AUROC} - 0.5|$ for **correct**; bold = best per row):

signal	quality	hotpot	trivia	musr
11_kl_wb_l	@30: 0.680	@34: 0.332	@33: 0.284	@34: 0.555
11_conf_w_l	@30: 0.353	@35: 0.731	@34: 0.653	@22: 0.428
11_margin_w_l	@30: 0.333	@35: 0.696	@34: 0.630	@19: 0.422
drift_l	@2: 0.374	@33: 0.397	@26: 0.370	@17: 0.531
11_argmax_agree_final_l	@34: 0.461	@34: 0.597	@23: 0.343	@34: 0.479

3.3 Interpretation

1. **The wrapper’s own confidence/margin predicts correctness** on the QA benchmarks (**conf_w/11_conf** AUROC 0.65–0.75). A confidence-thresholded gate is therefore viable: low confidence $\Rightarrow$ suppress the wrapper.
2. **The predictive layer is benchmark-dependent and often not the final one.** QuALITY’s strongest signal is logit-lens KL-vs-base at *layer 30* (**11_kl_wb@30**= 0.680), whereas Hotpot/Trivia peak at layers 34–35. This empirically validates the multi-layer hypothesis: a single-layer (output-only) gate would miss the QuALITY signal; a per-layer gate has the right information to read.
3. **Where there is no signal, the gate should simply close.** MuSR is at chance (max AUROC 0.58) and RULER is uniformly wrong (AUROC undefined). These are exactly the regimes where the wrapper must defer to context; a do-no-harm gate that closes there is correct, not a failure.
4. **Ensemble signals are stronger but costlier** (Phase 0: **n_baselines_agree** AUROC 0.83–0.90 on extractive benches), trading deployability for accuracy — a fallback if intrinsic signals prove insufficient under multi-seed.

4 The gate target and next steps

Mechanism. Implement the placeholder **gated combine** as an input-conditioned convex mix, optionally *per base layer ℓ* :

$$e'_t = (1 - g_\ell) e_t + g_\ell \tilde{m}_{t,\ell}, \quad g_\ell = \sigma(w_\ell^\top \phi_\ell(\text{signals}) + b_\ell), \quad b_\ell \ll 0, \quad (8)$$

so $g_\ell \approx 0$ at init (wrapper $\approx$ identity, base preserved; cf. LLaMA-Adapter zero-init gating, Flamingo tanh-gating, DISEL input-conditioned LoRA gating). The gate reads the signals the correlation study selects (likely late-layer **conf/11_conf** for correctness + mid-layer **11_kl_wb** for compressibility).

Training the gate. On `categorical_niah` + decoy items (irrelevant context $\Rightarrow$ gate should close, target = base `no_context` output) plus a do-no-harm KL regulariser $\text{KL}(p_{\text{wrap}} \| p_{\text{base}})$ on held-out OOD-like text. **Eval public-only**; success = Regime-A gain retained *and* `OURS_gated` $\geq$ `no_context` on Regime-B/C.

Pending. (i) Multi-seed bands (9 seeds) to firm up §3. (ii) Multivariate logistic probe for the gate ceiling. (iii) Multi-layer injection (only if the per-layer profile’s payoff over output-only justifies it).

Pointers

- Probe: `mem-embedding/scripts/probe_v2.py` (train+save, native-MC, 37-layer logit-lens/drift).
- Analysis: `scripts/signal_corr.py`; Phase 0 `scripts/signal_study.py`.
- Live ledger: `mem-embedding/summary/matrix.md` (v1 archived in `matrix-archive-2026-06-03.md`).
- Native metrics impl: `llm-infra/src/llm_infra/benchmark_metrics.py`.
- Runs: `runs/pv2-20260604-181915UTC` (seed 42), `runs/pv2-seed*-20260604-*UTC` (multi-seed).